

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE
(IOHE)**

ON

Rental House Price Prediction Using Machine Learning

submitted in partial fulfilment of the requirements for the award of degree of

**BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Rental House Price Prediction Using Machine Learning" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Swati Goel. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura
Date: May 2, 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The real estate rental market in India has witnessed exponential growth driven by rapid urbanization, migration of working professionals, and a growing middle-class population. In this scenario, accurately estimating rental prices has become a critical challenge, as traditional methods relying on real estate agents are often subjective, time-consuming, and unreliable. This project presents a comprehensive machine learning-based framework for predicting residential rental prices across six major Indian cities including Mumbai, Delhi, Bangalore, Kolkata, Chennai, and Hyderabad.

The study utilizes a publicly available dataset comprising 4,746 property listings with 12 structural and spatial features. Five regression algorithms were systematically implemented and evaluated: Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor. The dataset was rigorously preprocessed using Interquartile Range (IQR) based outlier removal, domain-specific feature engineering (extraction of floor number, total floors, and posting month), and label encoding of categorical variables.

Model performance was measured using Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). Hyperparameter optimization was performed using GridSearchCV with 5-fold cross-validation. Experimental results confirmed that ensemble methods significantly outperform simple regression approaches. The tuned Random Forest model achieved the highest R^2 score of 0.74, making it the most reliable model for real-world rental price prediction.

Keywords: Rental Price Prediction, Machine Learning, Random Forest, XGBoost, Regression, Feature Engineering, Real Estate Analytics, Indian Cities.

1. INTRODUCTION

India has witnessed a major transformation in its residential rental market over the last decade. Accelerating urbanization, large-scale migration of working professionals to metropolitan areas, and a rapidly expanding middle-class population have collectively driven demand for well-built rental housing across major cities. In cities like Mumbai, Delhi, Bangalore, Kolkata, Chennai, and Hyderabad, rental rates fluctuate due to a complex interplay of structural, locational, and socio-economic factors.

Traditional methods for estimating rental prices are predominantly based on the qualitative knowledge and personal judgment of real estate agents, making the process highly subjective and unreliable. These approaches are also time-consuming, not scalable, and fail to capture the nonlinear relationships between various property attributes and rental prices. The proliferation of online rental platforms has generated large volumes of structured property data, making it feasible to apply data-driven predictive modeling techniques.

Machine learning provides a powerful alternative to these conventional approaches, enabling models to learn complex patterns from historical data and accurately predict future values. Regression-based machine learning models have demonstrated strong performance in real estate price prediction tasks across multiple international studies. However, there is a notable lack of comprehensive comparative studies involving multiple regression algorithms applied to Indian rental data with thorough preprocessing and hyperparameter optimization.

This project addresses this gap by designing, implementing, and systematically comparing five ML regression models for rental price prediction. The key contributions include: (i) an exploratory analysis of Indian city rental data, (ii) a robust preprocessing workflow with outlier handling and domain-informed feature engineering, (iii) a systematic comparison of five ML models, (iv) hyperparameter tuning using GridSearchCV, and (v) a functional prediction interface that provides real-time rent estimates to end users.

2. METHODOLOGY

The project follows a structured and modular end-to-end machine learning pipeline, beginning from raw data acquisition and culminating in real-time rental price predictions. The overall system workflow is illustrated in Figure 1 below.

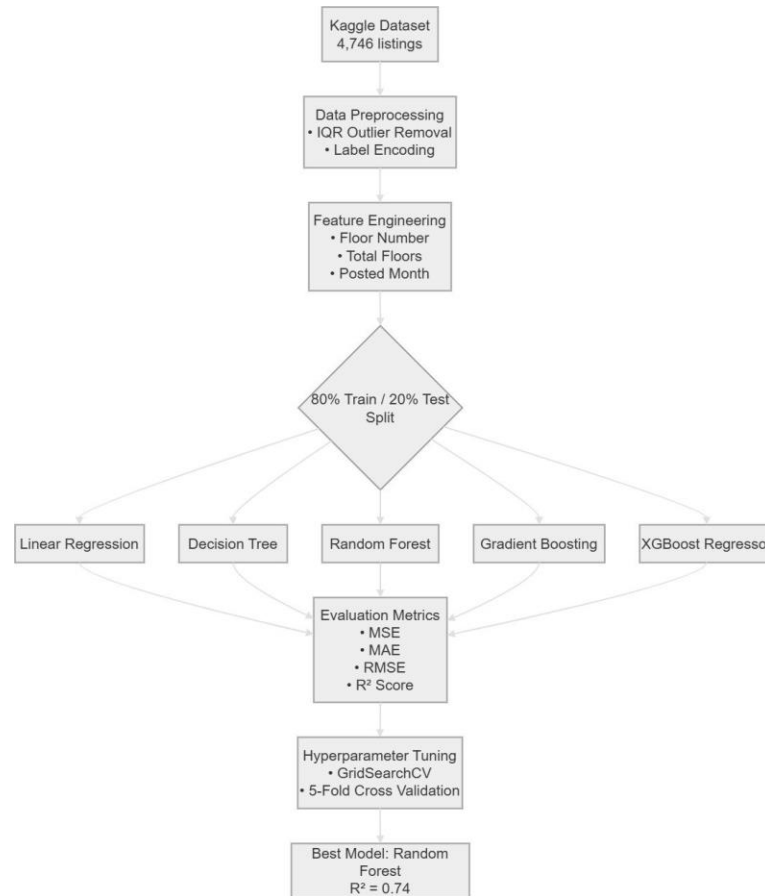


Figure 1: System Methodology Flowchart

2.1 Dataset Description

The dataset used in this study is the House Rent Dataset sourced from Kaggle, containing 4,746 residential property listings from six major Indian cities: Mumbai, Delhi, Bangalore, Kolkata, Chennai, and Hyderabad. Each record includes 12 features describing structural and spatial properties of the listing. The target variable, Rent, exhibits a heavily right-skewed distribution ranging from Rs. 1,200 to Rs. 35,00,000 per month, with a median of Rs. 16,000 and a mean of Rs. 34,993, the latter inflated by a few high-value outliers. Mumbai and Bangalore recorded the highest median rents

(approximately Rs. 32,000 and Rs. 22,000 respectively), while Kolkata had the lowest at around Rs. 10,000.

Feature	Type	Description
BHK	Numeric	Number of bedrooms, hall, and kitchen
Rent	Numeric	Monthly rental price in Indian Rupees (target variable)
Size	Numeric	Property area in square feet
Floor	Text	Floor number and total floors (e.g., '2 out of 5')
Area Type	Categorical	Super Area, Carpet Area, or Built Area
City	Categorical	One of six major Indian cities
Furnishing Status	Categorical	Furnished, Semi-Furnished, or Unfurnished
Tenant Preferred	Categorical	Bachelors, Family, or Bachelors/Family
Bathroom	Numeric	Number of bathrooms
Point of Contact	Categorical	Contact Owner, Agent, or Builder
Posted On	Date	Date when the listing was posted online

Table 1: Dataset Features and Descriptions

2.2 Data Preprocessing

The preprocessing stage followed a systematic sequence. Records with missing values were dropped first to ensure a clean dataset. Next, the IQR method was applied to remove outliers in the Rent column. Values outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ were eliminated, reducing the dataset to approximately 4,200 records. All categorical features including City, Area Type, Furnishing Status, Tenant Preferred, and Point of Contact were converted to numerical form using LabelEncoder from scikit-learn.



Figure 5: Feature Correlation Heatmap

2.3 Feature Engineering

The original Floor column contained complex text entries such as '2 out of 5'. Two new numerical features were engineered: FloorNumber and TotalFloors. Ground floor entries were assigned the value zero. The PostedOn date column was transformed into a PostedMonth indicator to capture seasonality effects. These engineered features were appended to the existing numerical attributes to create a richer and more informative feature set for the models.

2.4 Model Training and Evaluation

The processed dataset was split into 80% training and 20% testing subsets. Five regression algorithms were trained and evaluated: Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor. Models were assessed using MSE, MAE, RMSE, and R^2 score. The best performing model, Random Forest, was subsequently subjected to

hyperparameter tuning using GridSearchCV with 5-fold cross-validation to further optimize its generalization.

3. TOOLS AND TECHNOLOGIES

The entire project was developed using open-source tools and libraries within the Python ecosystem. The following tools and technologies were used:

3.1 Python 3.10

Python served as the primary programming language owing to its simplicity, versatility, and the richness of its data science ecosystem. Python 3.10 offered improved error messages, structural pattern matching, and enhanced type hints. Its wide adoption in the machine learning community and the availability of mature libraries made it the natural choice for implementing the entire pipeline from data preprocessing to model training and result visualization.

3.2 Pandas

Pandas is an open-source data manipulation and analysis library for Python. It was used extensively for loading the dataset from CSV format, inspecting data types and missing values, applying data transformations, computing statistical summaries, and performing feature engineering operations such as extracting floor numbers and posting months. Its DataFrame structure provided an intuitive tabular representation that significantly simplified data wrangling operations.

3.3 NumPy

NumPy (Numerical Python) is the fundamental package for scientific computing in Python. It provides support for large multi-dimensional arrays and matrices along with high-level mathematical functions. In this project, NumPy was used for array-level mathematical operations, computing IQR bounds for outlier detection, and supporting backend computations within scikit-learn and other ML libraries. Its vectorized operations greatly improved numerical computation efficiency.

3.4 Scikit-learn

Scikit-learn is the most widely used machine learning library for Python. It provides consistent APIs for data preprocessing, model training, evaluation, and hyperparameter tuning. In this project, scikit-learn was used to implement Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor. Additionally, LabelEncoder was used for categorical encoding, StandardScaler for feature scaling, GridSearchCV for hyperparameter optimization, and cross_val_score for 5-fold cross-validation.

3.5 XGBoost

XGBoost (Extreme Gradient Boosting) is an optimized and scalable implementation of gradient boosting. It incorporates L1 and L2 regularization to combat overfitting and supports parallel tree construction, making it significantly faster than standard gradient boosting. XGBoost was used as the fifth regression model in this comparative study. Its regularization capabilities and system-level optimizations make it particularly suitable for structured tabular datasets like the rental listing data.

3.6 Matplotlib and Seaborn

Matplotlib is Python's foundational plotting library, used to create bar charts comparing model performance metrics (R^2 , MAE, RMSE). Seaborn, built on top of Matplotlib, provides a higher-level interface for drawing attractive statistical graphics. It was used during Exploratory Data Analysis to visualize distributions of rental prices across cities, the feature correlation heatmap, and the impact of furnishing status on rent values.

3.7 Joblib

Joblib is a Python library providing utilities for lightweight pipelining and efficient serialization of Python objects. In this project, Joblib was used to save the trained Random Forest model to disk after hyperparameter tuning, enabling it to be reloaded at inference time without retraining. This is a critical step for deploying the model in a production environment or integrating it with a web interface such as Flask or Streamlit.

3.8 Jupyter Notebook

Jupyter Notebook was used as the interactive development environment throughout the project. It enabled iterative development, inline visualization of EDA charts, step-by-step model training, and easy documentation of the experimental process. Its cell-based execution model allowed experimentation with different preprocessing strategies and model configurations in a modular and reproducible manner.

4. IMPLEMENTATION

The implementation of the rental price prediction system followed a modular and structured pipeline approach. Each stage of the pipeline was developed independently to ensure maintainability, testability, and ease of future extension.

4.1 Data Loading and Exploration

The dataset was loaded using Pandas from a CSV file. An initial exploratory data analysis was performed to understand feature distributions, identify missing values, and detect outliers. Descriptive statistics such as mean, median, and standard deviation were computed for numerical features. City-wise rental price distributions were visualized using Seaborn. The correlation heatmap (Figure 5) was generated to understand inter-feature relationships, revealing strong positive correlations between BHK, Size, and Bathroom with the Rent column.

4.2 Preprocessing Pipeline

Missing value rows were removed first. The IQR method was then applied to the Rent column to eliminate extreme outliers. Categorical columns were encoded using LabelEncoder. The Floor column was parsed using string operations to extract FloorNumber and TotalFloors. The PostedOn date column was processed using Pandas datetime functions to extract the PostedMonth integer. After preprocessing, the cleaned dataset contained approximately 4,200 records ready for model training.

4.3 Model Training

The dataset was divided into training (80%) and testing (20%) sets using scikit-learn's `train_test_split` with a fixed random state for reproducibility. All five regression models were instantiated with initial configurations and trained on the training set. Linear Regression required StandardScaler to normalize

features prior to training. Decision Tree was configured with `max_depth=10` and `min_samples_split=20`. Random Forest and Gradient Boosting were both set to 200 estimators. XGBoost was configured with a learning rate of 0.1 and `max_depth` of 5.

4.4 Hyperparameter Tuning

GridSearchCV was applied to the Random Forest model over a parameter grid comprising `n_estimators`: {100, 200}, `max_depth`: {10, 15, None}, and `min_samples_split`: {5, 10}. The best configuration identified was `n_estimators=200`, `max_depth=15`, and `min_samples_split=5`. The 5-fold cross-validation R^2 score of the tuned model reached approximately 0.86, confirming its robustness across different data subsets. The final tuned model was saved to disk using Joblib for later inference.

4.5 Prediction Interface

A command-line prediction interface was implemented that accepts user inputs for property attributes such as city, size in square feet, BHK, furnishing status, floor number, and bathroom count. The saved model is loaded and the user-provided inputs are encoded and transformed to match the training feature format. The system then outputs the predicted monthly rent in Indian Rupees. This modular design allows straightforward integration with a web-based front-end using Flask or Streamlit in the future.

5. MAJOR FINDINGS / OUTCOMES / RESULTS

The experimental results revealed a clear performance hierarchy among the five regression models evaluated on the held-out test set. Table 2 presents the complete comparison. Linear Regression, while computationally efficient, recorded the weakest performance with an R^2 of 0.6162 and an RMSE of Rs. 8,578, confirming that rental prices exhibit highly nonlinear relationships that a simple linear model cannot capture adequately.

Ensemble models demonstrated substantially superior performance. The Random Forest Regressor achieved the highest R^2 score of 0.7430 with an RMSE of Rs. 7,020 and an MAE of Rs. 4,710, making it the best performing model overall. Gradient Boosting followed closely with $R^2 = 0.7339$, while XGBoost achieved $R^2 = 0.7225$. The 5-fold cross-validation R^2 of the tuned Random Forest was approximately 0.86, indicating strong generalization capability.

Model	MSE (Rs. ²)	MAE (Rs.)	RMSE (Rs.)	R ² Score
Linear Regression	73,588,755.59	6,291.20	8,578.39	0.6162
Decision Tree	60,676,783.04	5,197.17	7,789.53	0.6835
Random Forest	49,275,379.15	4,710.36	7,019.64	0.7430
Gradient Boosting	51,027,691.99	4,741.61	7,143.37	0.7339
XGBoost	53,214,972.00	4,850.39	7,294.86	0.7225

Table 2: Comparative Model Performance on Test Set

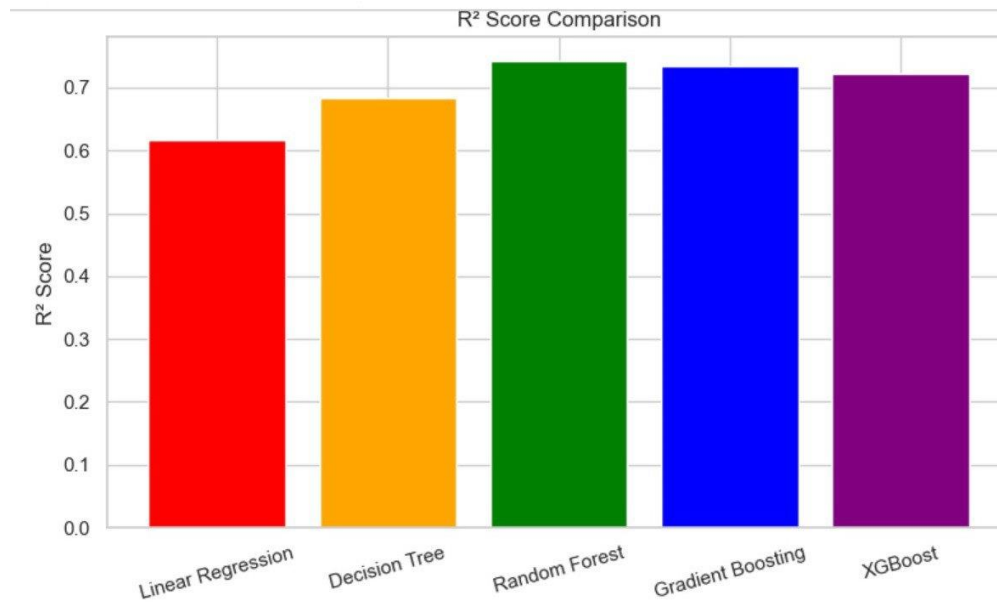


Figure 2: R² Score Comparison of All Models

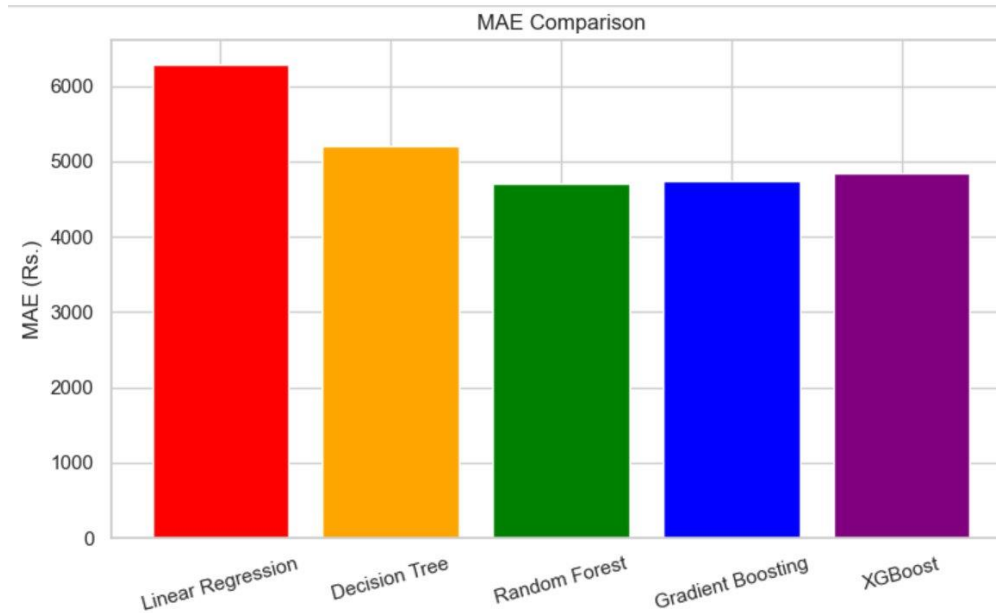


Figure 3: MAE Comparison of All Models

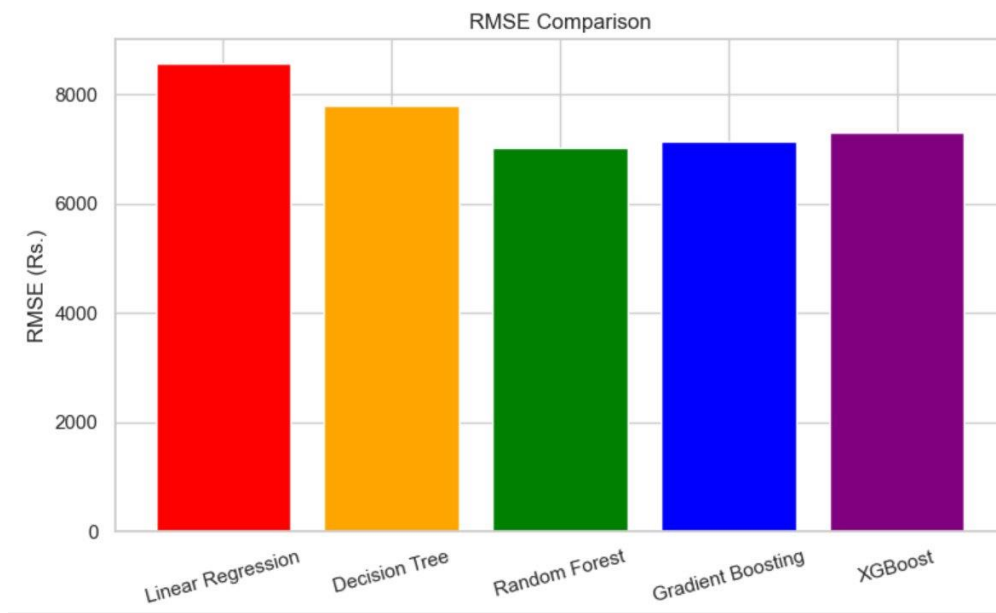


Figure 4: RMSE Comparison of All Models

Feature importance analysis from the Random Forest model revealed that City and Size together contributed over 45% of the predictive power, followed by BHK, Bathroom, and FloorNumber. Furnishing Status and Tenant Preferred contributed moderately, while PostedMonth had minimal

impact. These findings align well with real estate domain expertise, where location and property size are universally recognized as the dominant drivers of rental value.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully demonstrated the application of machine learning techniques for predicting residential rental prices in major Indian cities. A complete end-to-end pipeline was developed encompassing data preprocessing, feature engineering, model training, evaluation, and prediction. Among the five regression models evaluated, ensemble methods, particularly the Random Forest Regressor, significantly outperformed simpler approaches. The tuned Random Forest model achieved an R^2 of 0.74 on the test set and approximately 0.86 during cross-validation, confirming its robustness and generalizability.

The feature importance analysis validated real-world real estate knowledge by identifying city and property size as the dominant predictors of rental price. The modular software architecture ensures that individual components of the pipeline can be independently updated or replaced. The prediction interface provides a practical tool that tenants, landlords, and real estate professionals can use to obtain data-driven rental price estimates, thereby reducing dependence on subjective agent opinions.

6.2 Future Scope

Several meaningful directions can extend this work. Incorporating additional features such as proximity to schools, hospitals, public transport hubs, and commercial centers could substantially improve prediction accuracy, as environmental and locational amenities are known determinants of rental value. Expanding the dataset to include more Indian cities and more recent listings would enhance the model's geographic and temporal generalizability.

Deep learning architectures such as feed-forward neural networks or attention-based transformer models could be explored for further performance gains. Developing and deploying a fully functional web application using Flask or Streamlit would make the system accessible to a broader user base.

Additionally, integration of real-time listing data via web scraping APIs could enable the model to continuously retrain on fresh data, keeping predictions current with evolving market conditions.

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